LESSON 2.4

# Represent and Describe Data in a Box Plot

**Start here** · Words, worked example, and sentence starters

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can make and read a box plot to summarize a data set.

**WHY THIS MATTERS** The basketball league wants to compare scoring across teams. Here are the points scored by each player on the Wildcats in their last game: 4, 6, 8, 10, 12, 14, 15, 18, 22, 24, 28. The league needs a quick visual that shows the spread, the middle 50%, and any high or low scorers. Time to build a box plot!

START HERE

## Words for this lesson

Read each word before you begin. Say it out loud.

| WORD | WHAT IT MEANS | EXAMPLE |
|---|---|---|
| **Box Plot** *Spanish: Diagrama de caja* | A graph that shows how data is spread using five key numbers. | Box with two whiskers |
| **Median** *Spanish: Mediana* | The middle number when you put them in order. | Data: 10, 15, 20, 25, 30 → median is 20 (the 3rd value) |
| **Quartile** *Spanish: Cuartil* | Numbers that split the data into four equal parts. | Q1, median, Q3 |
| **Interquartile Range** *Spanish: Rango intercuartílico* | The spread of the middle half of the data (Q3 − Q1). | If Q1 = 12 and Q3 = 20, then IQR = 20 − 12 = 8 — the middle 50% spans 8 units |
| **Data distribution** *Spanish: Distribución de datos* | How the data looks: where it sits and how spread out it is. | A box plot where the median is centered in the box shows symmetric distribution |
| **Variability** *Spanish: Variabilidad* | How spread out the numbers are. | Small IQR = low variability in the middle 50%. Large IQR = high variability |

HOW IT WORKS

## Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

**Times (min): 14, 5, 20, 9, 23, 11, 16.**

1. Order: 5, 9, 11, 14, 16, 20, 23. Median = 14.
2. Lower half 5, 9, 11: Q1 = 9. Upper half 16, 20, 23: Q3 = 20.
3. Box from 9 to 20, line at 14, whiskers to 5 and 23.

**Answer:** 5, 9, 14, 20, 23

NOW YOU TRY

## Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

**New, smaller data set: 2, 5, 5, 8, 11. First, what is the median (the middle value)?**

1. _____
2. _____
3. _____
4. _____

**Answer:** _____

SAY IT AND WRITE IT

## Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- In order the values are _____ .
- The median is _____ , so the lower half is _____ .
- Quartile 1 is _____ and quartile 3 is _____ .
- The box stretches from _____ to _____ , which holds _____ of the data.

**Word bank** box plot | quartile | median | minimum | maximum | whisker | box | order | lower half | upper half | middle | spread

WATCH OUT!

## A common mistake

A common mistake in Display Data: Box Plots is treating a quartile as just one data point instead of the median of a half

LESSON 2.4

# Represent and Describe Data in a Box Plot

**Version A** · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can make and read a box plot to summarize a data set.

**WHY THIS MATTERS** The basketball league wants to compare scoring across teams. Here are the points scored by each player on the Wildcats in their last game: 4, 6, 8, 10, 12, 14, 15, 18, 22, 24, 28. The league needs a quick visual that shows the spread, the middle 50%, and any high or low scorers. Time to build a box plot!

## PART 1 Understand the idea

models and meaning

**1** **Circle the letter of the best answer. Show how you know.**

On a box plot, what does the line inside the box represent?

(A) The median
(B) The mean
(C) The mode
(D) The range

**HINT** Picture a box plot: a box with whiskers, and a line cutting through the box. That line marks the CENTER of the data.

SHOW YOUR WORK

**2** **Write the name of the correct group on each line.**

Order the steps for making a box plot.

**Groups:** Steps (in order)

_____ Step 1: Order the data from least to greatest

_____ Step 2: Find the median (middle value)

_____ Step 3: Find Q1 (median of the lower half)

_____ Step 4: Find Q3 (median of the upper half)

_____ Step 5: Draw the box from Q1 to Q3 with a line at the median, and whiskers to min and max

**HINT** Always order the data first before finding any measures.

## PART 2 Apply it

real-world problems

**3** **Circle the letter of the best answer. Show how you know.**

A box plot shows: Min = 10, Q1 = 15, Median = 20, Q3 = 28, Max = 35. What is the interquartile range (IQR)?

(A) 10
(B) 13
(C) 20
(D) 25

**HINT** You're given five numbers, but IQR only uses two of them: Q1 and Q3. Ignore the min, median, and max.

**Put the data in order** _____

**WORK**

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

**4** **Circle the letter of the best answer. Show how you know.**

What percentage of data falls between Q1 and Q3 on a box plot?

(A) 25%
(B) 50%
(C) 75%
(D) 100%

**HINT** Quartiles cut the data into 4 equal parts — 'quart' like 'quarter.' Each part holds the same share of the data.

**Put the data in order** _____

**WORK**

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

**5** **Circle the letter of the best answer. Show how you know.**

Find the range of: 14, 22, 8, 31, 17

(A) 8 (B) 18

(C) 23 (D) 31

**HINT** Range = how far the data stretches: the gap between the biggest and smallest values only. The middle values don't matter.

**Put the data in order** _____

**WORK**

1. **Order** *Smallest to largest.*
2. **Count** *How many values?*
3. **Locate** *Middle, or add and divide.*
4. **Answer** *Say which measure it is.*

## PART 3 Explain and correct

reasoning

**6** **Find the mistake. Explain it, then show the correct work.**

### The Added-Quartiles IQR Mistake

1. **Box plot values** Bobcats' box plot: Min = 10, Q1 = 18, Median = 24, Q3 = 30, Max = 40
2. **Goal** Find the interquartile range (IQR).
3. **Student's work** IQR = Q3 + Q1 = 30 + 18 = 48

**Which step has the mistake? Explain what went wrong, then show the correct work.**

**HINT** IQR stands for InterQuartile RANGE — a range is found by subtracting, not adding.

**CORRECT WORK**

**Correct answer:** _____

EXPLAIN YOUR THINKING

**The IQR is Q3 − Q1 = 26 − 10 = 16. What does this tell you about the middle 50% of scorers?**

**SENTENCE STARTER** The middle 50% of players scored between ___ and ___ points. The IQR of ___ means the middle half of the data is spread across ___ points. This tells me ___.

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 2.4

# Represent and Describe Data in a Box Plot

**Version B** · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can make and read a box plot to summarize a data set.

**WHY THIS MATTERS** The basketball league wants to compare scoring across teams. Here are the points scored by each player on the Wildcats in their last game: 4, 6, 8, 10, 12, 14, 15, 18, 22, 24, 28. The league needs a quick visual that shows the spread, the middle 50%, and any high or low scorers. Time to build a box plot!

## PART 1 Understand the idea

models and meaning

**1** **Complete the table. Show how you found each value.**

Complete the table.

| Measure | Value |
|---|---|
| Data: 5, 8, 12, 15, 18, 20, 25, 30, 35 | |
| Minimum | |
| Q1 (median of lower half: 5, 8, 12, 15) | |
| Median | |
| Q3 (median of upper half: 20, 25, 30, 35) | |
| Maximum | |
| IQR (Q3 − Q1) | |

SHOW YOUR WORK

**2** **Write the letter of the matching item on each line.**

Match each box plot term to its meaning.

| | |
|---|---|
| _____ Minimum | (A) Smallest value in the data set |
| _____ Q1 | (B) Median of the lower half |
| _____ Median | (C) Middle value of the entire data set |
| _____ Q3 | (D) Median of the upper half |
| _____ IQR | (E) Q3 − Q1 (spread of middle 50%) |
| _____ Maximum | (F) Largest value in the data set |

**3** **Mark and label each value on the number line.**

Data: 3, 5, 7, 9, 11, 13, 15. Place the five-number summary on the number line.




SHOW YOUR THINKING

**4** **Circle the letter of the best answer. Show how you know.**

Two box plots show: Team A has IQR = 8, Team B has IQR = 22. What does this tell you?

(A) Team A's middle 50% is more consistent; Team B's is more spread out

(B) Team B is a better team

(C) Team A has a higher median

(D) The IQRs don't tell you about consistency

SHOW YOUR WORK

## PART 2 Explain and correct

reasoning

**5** **Find the mistake. Explain it, then show the correct work.**

### The Quartile-Is-One-Value Mistake

1. **Data (ordered)** Eagles' game scores: 4, 8, 10, 14, 18, 22, 26, 30 (8 values)
2. **Median** With 8 values, Median = (14 + 18) ÷ 2 = 16 ✓
3. **Q1** Lower half = 4, 8, 10, 14 → Q1 = 10
4. **Q3** Upper half = 18, 22, 26, 30 → Q3 = (22 + 26) ÷ 2 = 24

**Which step has the mistake? Explain what went wrong, then show the correct work.**

CORRECT WORK

**Correct answer:** _____

EXPLAIN YOUR THINKING

**The IQR is Q3 − Q1 = 26 − 10 = 16. What does this tell you about the middle 50% of scorers?**

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 2.4

# Represent and Describe Data in a Box Plot

**Challenge** · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

**Name** _____ **Date** _____ **Period** _____

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**WHY THIS MATTERS** The basketball league wants to compare scoring across teams. Here are the points scored by each player on the Wildcats in their last game: 4, 6, 8, 10, 12, 14, 15, 18, 22, 24, 28. The league needs a quick visual that shows the spread, the middle 50%, and any high or low scorers. Time to build a box plot!

## PART 1 Understand the idea

models and meaning

**1** **Write the letter of the matching item on each line.**

Match each box plot feature with what it tells you about the data.

_____ Long right whisker, short left whisker

_____ Median line centered in the box

_____ Very small box (small IQR)

_____ Very large box (large IQR)

(A) Skewed right — some high values stretch far from the box

(B) Approximately symmetric in the middle 50%

(C) Middle 50% is tightly clustered — consistent

(D) Middle 50% is widely spread — high variability

## PART 2 Explain and correct

reasoning

**2** **Find the mistake. Explain it, then show the correct work.**

### Find the Box Plot Error

1. **Data** 3, 7, 9, 12, 15, 18, 21
2. **Min and Max** Min = 3, Max = 21
3. **Median** Middle value = 12 ✓
4. **Q1 and Q3** Q1 = 9, Q3 = 18

**Which step has the mistake? Explain what went wrong, then show the correct work.**

CORRECT WORK

**Correct answer:** _____

**3** **Answer in complete sentences.**

Two basketball teams played 9 games each. Team A: 40, 42, 44, 46, 48, 50, 52, 54, 56. Team B: 30, 35, 40, 45, 48, 50, 55, 60, 65. Find the five-number summary and IQR for each team. Which team is more consistent? Which has a wider spread in the middle 50%?

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

**The IQR is Q3 − Q1 = 26 − 10 = 16. What does this tell you about the middle 50% of scorers?**

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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